

Code-Layout, Readability and Reusability

Date	14 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	
2	Proper File Structure	Files and folders are logically organized	Yes	Proper indentation maintained throughout the project for better readability.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Source files, templates, static assets, model, and dataset are organized logically.
4	Function / Method Names	Functions are descriptively named	Yes	Variables and functions use descriptive names representing their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Functions are clearly named according to their responsibilities.
6	Modular Design	Code is split into reusable functions/modules	Yes	Important sections are documented; additional comments can improve maintainability.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate logic is minimized to improve maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation and error handling are implemented; more comprehensive exception handling can be added.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Input Validation Module	Python, Flask	User input processing	High

2	Crop Prediction Module	Scikit-learn, Joblib	Machine learning prediction	High
3	Web Interface	HTML, CSS, JavaScript	User interaction pages	Medium
4	Dataset Loader	Pandas	Data preprocessing and model loading	Medium
5	Prediction Result Display	Flask Templates (HTML)	Displaying recommended crop	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Project files are logically organized with a clear separation of frontend, backend, ML model, and dataset.
Readability	4	The code uses meaningful variable and function names with consistent formatting, making it easy to understand.
Reusability	4	The prediction logic and input validation can be reused in future agricultural applications or extended with new features.
Documentation / Comments	3	Basic documentation is available. Additional inline comments and developer documentation would further improve maintainability.
Overall Score	4/5	The project follows good coding practices, maintains a clean structure, and provides a reusable machine learning solution for crop recommendation.